

Braillix - Assembly Bible & Wiring Guide

[!IMPORTANT] **CURRENT RELEASE GATE - 2026-08-19:** Do not assemble the full printed mechanism from this guide yet. The production cam has an invalid 3.5mm socket for the measured 7.5mm shaft, and the complete stack remains on HOLD. Print only the files in `printing/gcode_kobra_neo_checked` and follow `PRINT_RELEASE_2026-08-19.md` plus `MECHANISM_BENCH_TEST.md`. Resume final assembly only after a later release explicitly removes this gate.

Follow this guide sequentially to build your first single-cell prototype.

SECTION 1: CRITICAL SAFETY CHECKS (DO THIS FIRST)

[!CAUTION] **BARREL JACK POLARITY CHECK** The photographed inline female pigtail has red/black wires, but colour is not proof. If you wire 5V backwards, the ULN2003 chip can burn out. 1. Plug the 5V adapter into the wall. 2. Plug the adapter's round male plug into the black inline female pigtail. Keep both stripped wires disconnected from the circuit. 3. Turn on your Multimeter and set it to **DC Voltage (V DC)**. 4. Touch the RED probe to the exposed red wire and the BLACK probe to the exposed black wire. 5. If the screen reads approximately +5.0V, red is **POSITIVE (+)** and black is **GND (-)**. 6. If it reads approximately -5.0V, the pigtail colours are reversed. Do not connect it until the wires are relabelled. 7. Disconnect wall power before attaching either wire to the breadboard.

SECTION 2: WIRING DIAGRAM

We are wiring the **ESP32**, **ULN2003**, and **Hall Sensor** together.

```
[5V/3A Adapter]
|
+---> (+) ---> [BREADBOARD 5V RAIL] ---> ULN2003 (+) power pin
|
+---> (-) ---> [BREADBOARD GND RAIL] ---> ULN2003 (-) power pin

[ESP32 (Powered by USB)]
GND -----> [BREADBOARD GND RAIL] (Crucial: Common Ground!)
3V3 -----> Hall Sensor VCC
GND -----> Hall Sensor GND
D34 <----- Hall Sensor AO (Analog Out)
D18 -----> ULN2003 IN1
D19 -----> ULN2003 IN2
D21 -----> ULN2003 IN3
D22 -----> ULN2003 IN4

[ULN2003 Output]
White JST Socket -----> 28BYJ-48 Stepper Motor
```

Kill-Shots to Avoid:

- 1 **Never** connect the 5V rail to the Hall Sensor VCC. ESP32 pins are not 5V tolerant. Use 3V3.
- 2 **Never** connect the 5V adapter rail to the ESP32 VIN pin while the USB cable is plugged into your laptop. Power the motor from the adapter, and the ESP32 from USB. Only link their grounds.

SECTION 3: CURRENT PRE-ASSEMBLY VALIDATION SEQUENCE

Phase A: PETG fit evidence

- 1 Print `motor_cam_socket_coupon.gcode`; test the unpowered shaft from 3.1 to 3.3mm and record the smallest removable fit. Never hammer, heat, power, glue, or cut the shaft during this test.
- 1 Print `base_interface_coupon.gcode`. Inspect the dia 29 motor-seat roof for sag, test the actual M4 screw in the 3.3/3.4/3.5mm 2mm-deep pilots, and test the M2.5 bolt in the 2.9/3.0/3.1mm dia 6x12 standoffs. Do not force a bore that whitens or splits.
- 1 Print the remaining required coupons in `PRINT_RELEASE_2026-08-19.md` and record every selected fit before changing a production dimension.

Phase B: electronics bring-up outside the enclosure

- 1 Keep the ESP32 on USB and the motor on the 5V adapter; connect grounds.
- 2 Power the bare Hall sensor/module from ESP32 3V3, not 5V, and use AO for the initial test.

3 Verify motor direction, homing response, and the already measured red-positive pigtail polarity on the breadboard. Do not install electronics in an unreleased enclosure.

Phase C: cam/linkage/spring bench evidence

- 1 Print `cam_linkage_test_fixture.gcode` in PETG.
- 2 Obtain the four-part `cad/stl/cam_linkage_test_resin_set.stl` in resin plus the actual 2mm OD spring. Do not use PETG linkages or a pen spring.
- 1 Follow `MECHANISM_BENCH_TEST.md` for dot-5 and dot-6 dry travel, 50-cycle return, dwell, damage, height, and torque tests. The fixture is evidence equipment, not the final enclosure.

Phase D: final assembly - intentionally blocked

Do not press the current production cam onto the motor, trim the shaft, glue the Hall sensor, glue magnets, install heat-set inserts, or glue the resin dot insert yet. Those actions either hide a failed fit or are difficult to reverse. A later validated release must provide the corrected cam, base/Hall stack, enclosure height, selected coupon dimensions, and final assembly sequence.

SECTION 4: BEGINNER SOLDERING MISTAKES

If you are doing the flat-soldering for the ULN2003:

- **Cold Joint:** Looks like a dull, grey blob that doesn't stick to the pad. *Fix: Apply a tiny bit of flux, heat the pin and wire simultaneously for 2 seconds, and reapply a dab of solder.*
- **Bridging:** Solder spills over and connects two pins (e.g., IN1 and IN2). This will break the motor steps. *Fix: Heat the bridge and swipe your iron away quickly, or use desoldering wick to soak up the excess.*
- **Melted Insulation:** Holding the iron on the wire too long melts the plastic jacket. *Fix: Tin the wire and pin separately first (apply solder to them individually), then just touch them together with the iron for 1 second to fuse them.*